Honors Biology Unit 3A Enzymes and ATP Script

Slides 1 through 16

1. Today we are beginning our study of metabolism, enzymes, and ATP. These ideas are closely connected because cells depend on chemical reactions to build molecules, break molecules apart, and transfer energy. Enzymes control the rates of many of those reactions, while ATP provides a readily usable form of energy for cellular work. As we move through this unit, keep asking two questions: What is changing in the reaction, and where is the energy going?
2. Metabolism refers to the complete collection of chemical reactions occurring within an organism. Some reactions build molecules, while others break them down. Catabolism specifically describes reactions that break complex molecules into simpler compounds. Digestion is a familiar example because it breaks large food molecules into smaller molecules that cells can use. Later, we will contrast catabolic reactions with anabolic reactions, which build larger molecules from smaller components.
3. A chemical reaction changes one set of substances, called reactants, into another set of substances, called products. The equation on the slide shows sodium reacting with chlorine to form sodium chloride. According to the law of conservation of matter, the reaction cannot create or destroy atoms. Look carefully at the equation and consider this question: If the atoms are still present before and after the reaction, what actually happens to them during the process?
4. During a chemical reaction, atoms are rearranged. Existing chemical bonds break, new bonds form, and the same atoms become organized into different substances. In this example, sodium and chlorine atoms combine to form sodium chloride. Notice that the number of each type of atom remains consistent on both sides of the equation. Balancing an equation reflects this conservation of matter by accounting for every atom involved in the reaction.
5. Chemical reactions also involve energy changes. In an exergonic reaction, the reactants contain more free energy than the products, so the reaction releases energy to the surroundings. In an endergonic reaction, the products contain more free energy than the reactants, so the reaction requires an energy input. Cells often connect these two types of reactions so that energy released by one process can help drive another process that requires energy.
6. These graphs show the energy differences between exergonic and endergonic reactions. On the exergonic graph, the products end with less free energy than the reactants, so energy has been released. On the endergonic graph, the products end with more free energy, which means the reaction required an input. Catabolic reactions generally break molecules down and release energy. Anabolic reactions build larger molecules and generally require energy, so identify which graph best represents each process.
7. Many reactions that cells need would occur far too slowly without enzymes. Every reaction must overcome an initial energy barrier called activation energy. A catalyst increases the reaction rate by lowering that barrier. Enzymes are biological catalysts, and most are proteins. They make reactions easier to begin, but they do not change the overall amount of energy stored in the reactants or products. They provide a more efficient reaction pathway.
8. This estimate about hemoglobin and chlorophyll illustrates why enzymes are essential. Without enzymes, reactions needed to build biologically important molecules could take an impractical amount of time. Living cells cannot wait years, much less billions of years, for necessary reactions to occur. Enzymes allow metabolism to proceed rapidly enough to support growth, repair, energy transfer, and the other processes required for life.

9. The reactant that an enzyme acts upon is called its substrate. The substrate binds to a particular region of the enzyme called the active site. While the substrate is bound, the enzyme and substrate form an enzyme-substrate complex. The enzyme then helps convert the substrate into one or more products. Pay attention to the importance of shape in the diagram because the relationship between the substrate and active site determines whether binding can occur.
10. The lock-and-key model helps explain enzyme specificity. Just as a particular key fits a particular lock, an enzyme's active site has a shape and chemical properties that allow certain substrates to bind. This model is useful, although real enzymes may change shape slightly as a substrate binds. After the reaction, the products leave, and the enzyme remains available to work again. Enzymes participate in reactions without being permanently changed or consumed.
11. Enzymes can catalyze both catabolic and anabolic reactions. In a catabolic reaction, the enzyme helps break a larger substrate into smaller products, and the process generally releases energy. In an anabolic reaction, the enzyme helps join smaller substrates to form a larger product, and the process generally requires or stores energy. In either case, the enzyme lowers the activation energy. It does not determine whether the overall reaction releases or requires energy.
12. Some enzymes also have an allosteric site, which is a secondary binding site located away from the active site. In allosteric activation, an effector molecule binds to this site and changes the enzyme's shape. That change makes the active site better able to bind the substrate. This provides cells with a way to regulate enzyme activity. A signal at one location on the enzyme can influence what happens at the active site.
13. Enzyme activity depends on conditions that preserve the enzyme's three-dimensional shape. Temperature and pH can change molecular interactions within the protein. If conditions move too far outside the enzyme's functional range, the protein may denature and lose the shape of its active site. Substrate concentration also affects reaction rate. Adding more substrate increases the rate until nearly all active sites remain occupied. At that point, the enzymes are saturated, and the reaction rate plateaus.
14. The pH scale describes how acidic or basic a solution is by reflecting the concentration of hydrogen ions. A pH of 7 is neutral, lower values are acidic, and higher values are basic. Each enzyme functions best within a particular pH range because pH changes can alter the bonds that maintain protein shape. Many cellular environments remain near neutral, although some organelles and body systems function under more acidic or basic conditions. Moving too far from an enzyme's optimum can reduce its activity or cause denaturation.
15. Noncompetitive inhibition occurs when an inhibitor binds to an allosteric site rather than the active site. The inhibitor changes the enzyme's shape, which changes the active site and prevents the substrate from binding effectively. Because the inhibitor does not compete directly for the active site, adding more substrate usually does not overcome this form of inhibition. The key idea is that binding at one part of the enzyme can change the function of another part.
16. Competitive inhibition occurs when an inhibitor competes with the substrate for the active site. If the inhibitor occupies that site, the substrate cannot bind, and the enzyme cannot catalyze the reaction at that moment. Unlike noncompetitive inhibition, increasing substrate concentration may reduce the inhibitor's effect because more substrate molecules compete for the active site. Compare the two inhibition mechanisms by focusing on where the inhibitor binds and how that binding affects enzyme function.